

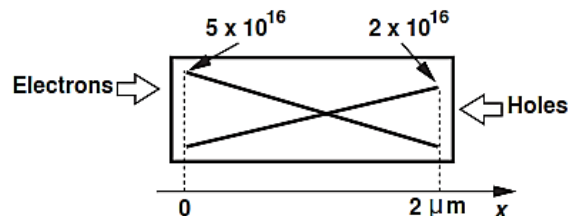
DEPARTMENT OF ELECTRONICS ENGINEERING
Indian Institute of Technology (Indian School of Mines), Dhanbad
Dhanbad, Jharkhand- 826004, India

SUBJECT: ELECTRONICS ENGINEERING (ECC 12101)

CLASS: B. TECH 2ND SEMESTER, SEC.-A,B,C,D

ASSIGNMENT I

1. Determine the probability that an energy level $3kT$ above the Fermi energy is occupied by an electron. Let $T = 300\text{ K}$.
2. A Si sample is doped with 10^{17} As atoms/cm³. What is the equilibrium hole concentration p_0 at 300 K? Where is E_F relative to E_i ? ($n_i = 9.65 \times 10^9$)
3. An n-type silicon bar 0.1 m long and $100\text{ }\mu\text{m}^2$ in cross-sectional area has a majority carrier concentration of $5 \times 10^{20}/\text{m}^3$ and the carrier mobility is $0.13\text{ m}^2/\text{V-s}$ at 300K. If the charge of an electron is 1.6×10^{-19} coulomb, then calculate the resistance of the bar.
4. Determine the density of electrons in silicon at $T = 300\text{ K}$ (room temperature) and $T = 600\text{ K}$. [$E_g = 1.12\text{ eV}$, $B = (5.2 \times 10^{15})^2$]
5. A uniform piece of n-type of silicon that is $1\text{ }\mu\text{m}$ long senses a voltage of 1 V . Determine the velocity of the electrons.
6. The electron concentration in a sample of uniformly doped n-type silicon at 300K varies linearly from $10^{17}/\text{cm}^3$ at $x=0$ to $6 \times 10^{16}/\text{cm}^3$ at $x = 2\text{ }\mu\text{m}$. Assume a situation that electrons are supplied to keep this concentration gradient constant with time. If electronic charge is 1.6×10^{-19} coulomb and the diffusion constant $D_n = 35\text{ cm}^2/\text{s}$, calculate the current density in the silicon, if no electric field is present.
7. The resistivity of a uniformly doped n-type silicon sample is $0.5\text{ }\Omega\text{-cm}$. If the electron mobility (μ_n) is $1250\text{ cm}^2/\text{V-sec}$ and the charge of an electron is 1.6×10^{-19} coulomb, calculate the donor impurity concentration (N_D) in the sample.
8. The mobilities of electron and hole in silicon are respectively $1350\text{ }\frac{\text{cm}^2}{\text{V-sec}}$ and $480\text{ }\frac{\text{cm}^2}{\text{V-sec}}$ and intrinsic carrier concentration of silicon is $1.5 \times 10^{10}/\text{cm}^3$. Then find the majority and minority concentrations of a p-type semiconductor which is having resistivity $10\text{ }\Omega\text{-cm}$.
9. The resistance of number 18 copper wire (diameter 1.03 mm) is $0.0214\text{ }\Omega/\text{m}$. The concentration of free electron is $8.4 \times 10^{28}\text{ electron}/\text{cm}^3$. If it is carrying a current of 2 A . Find drift velocity and mobility of electron.
10. Figure shows a p-type bar of Silicon that is subjected to electron injection from the left and hole injection from the right. Determine the total current flowing through the device if the cross-section area is $1\text{ }\mu\text{m} \times 1\text{ }\mu\text{m}$.



11. A n-type piece of Silicon with a length of $0.1\text{ }\mu\text{m}$ and a cross section area of $0.05\text{ }\mu\text{m} \times 0.05\text{ }\mu\text{m}$ sustains a voltage difference of 1 V . If the doping level is 10^{17} cm^{-3} , calculate the total current flowing through the device at $T = 300\text{ K}$.